Exp. 3
23/7/26

Understanding the cisco packet Tracer Environment

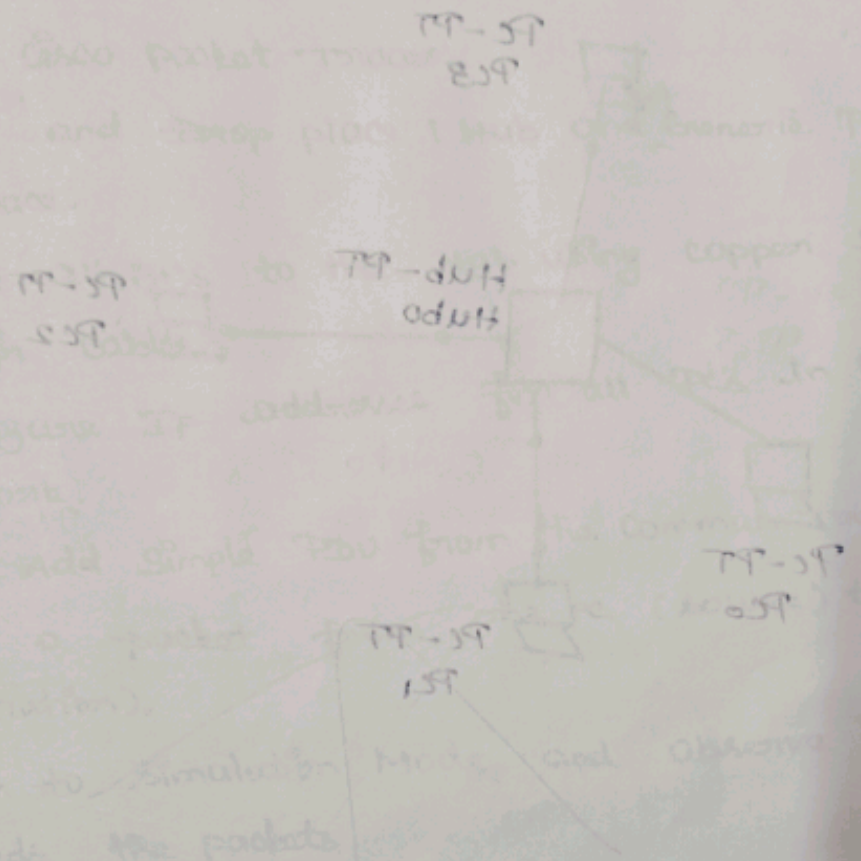
AIM:

To understand the cisco packet Tracer environment and design a simple computer network using its interface and networking components.

Steps:

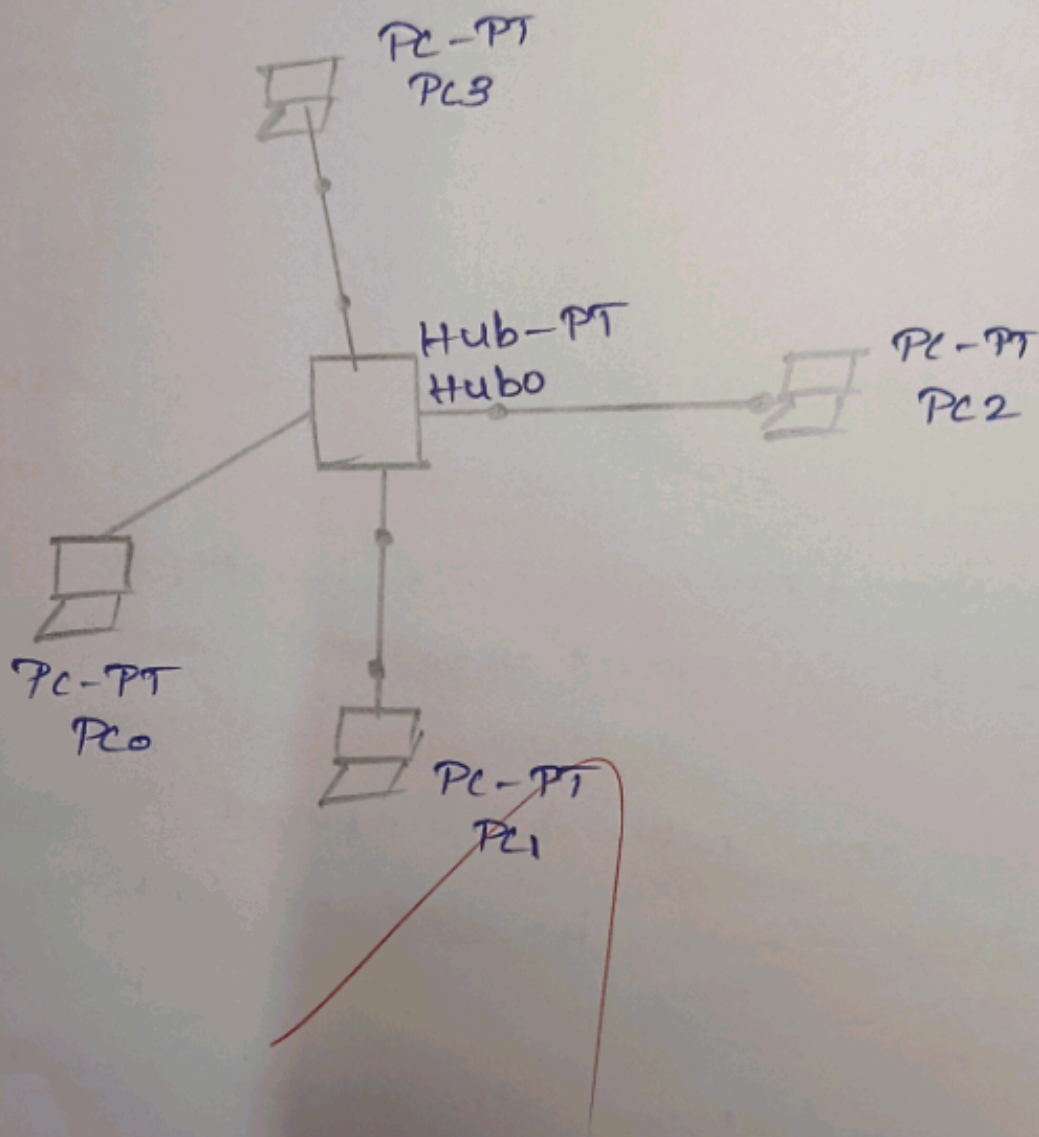
* opens the cisco packet Tracer application.
* Observe the user interface, including the Menu Bar, Main Toolbar, Workspace, Comman Tools, Network Component Box, and Realtime/Simulation tabs.
* From the network component Box, select the required network devices such as PCs, Hub and Switch.
* Drag and Drop the devices into the workspace.
* Connect the devices into the using Copper cables.
* Configure IP addresses for the PCs using the Desktop → IP configuration option.
* Verify that all links are active (green LED).
* Use the Add Simple PDU tool to send a packet between two PCs.
* Observe packet transmission in Realtime Mode
Mode.

* Save the network topology.



RESULT:

The Cisco Packet Tracer environment was successfully explored, and a simple network was designed. The interface components, device connections, IP configuration and packet transmission were successfully understood using both realtime and simulation modes.


Pc-PT
Pc3
Hub-PT
Hub0
Pc-PT
Pc2
Pc-PT
Pc0
Pc-PT
Pc1

Experiment 1 : HUB

AIM:

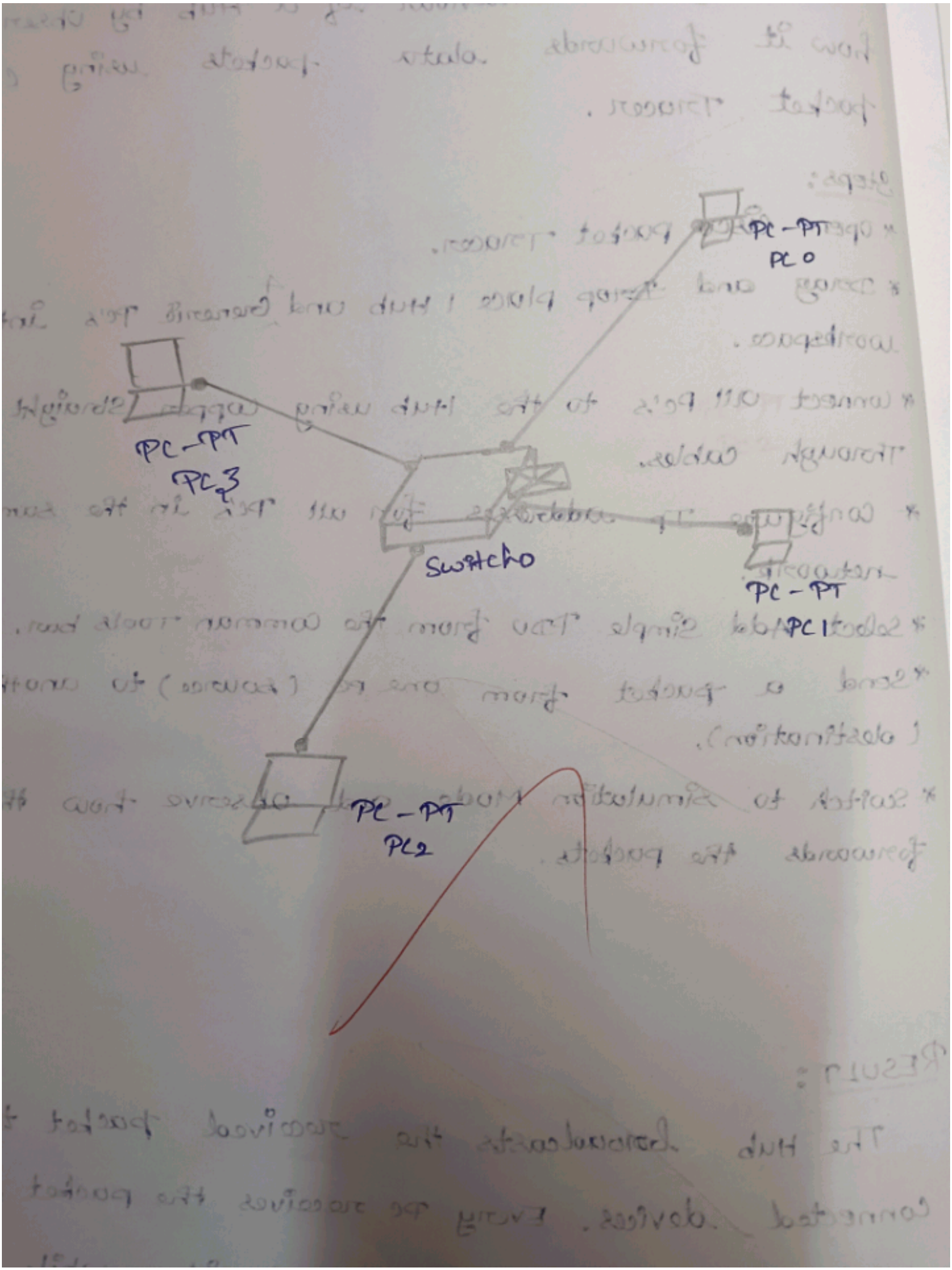
To study the behaviour of a Hub by observing how it forwards data packets using cisco packet Tracer.

Steps:

* Open Cisco packet Tracer.
* Drag and Drop place 1 Hub and 4 Generic PC's into the workspace.
* Connect all PC's to the Hub using copper Straight - Through cables.
* Configure IP addresses for all PC's in the same network.
* Select Add Simple PDU from the Comman Tools bar.
* Send a packet from one PC (source) to another PC (destination).
* Switch to Simulation Mode and Observe how the Hub forwards the packets.

RESULT:

The Hub broadcasts the received packet to all connected devices. Every PC receives the packet, but only the destination PC accepts it, while the others discard it.


PC-PT
PC0
PC-PT
PC3
Switch0
PC-PT
PC1
PC-PT
PC2

EXPERIMENT 2: SWITCH

AIM:

To study the behavior of switch by observing how it forwards data packets using Cisco packet Tracer

STEPS:

* Opens Cisco Packet Tracer.
* Drag and Place 1 switch and 4 Generic PCs into the workspace.
* Connect all PC's to the switch using copper Straight Through cables.
* Configure IP's addresses for all PC's in the same network.
* Select Add Simple PDU from the comman Tools box.
* Send a packet from one PC (source) to another PC (destination).
* Switch to Simulation Mode and observe how the switch forwards the packet.

RESULT:

The switch forwards the packets only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and communication efficiency.